

BAZOORA

Product Requirements Document

Version 1.0 • June 2026

Digital Waste Management Platform

Progressive Web Application (PWA)

1. Product Overview

Bazoor is a Progressive Web Application (PWA) that digitizes waste management and collection coordination for local communities. It connects Local Government Units (LGUs), private hauling organizations, businesses, and residents on a single platform — replacing fragmented, manual systems with real-time visibility, structured scheduling, and digital communication.

1.1 Problem Statement

Communities relying on manual waste management face recurring challenges:

- Delayed and inconsistent garbage collection schedules
- No real-time visibility into truck locations or route progress
- No formal digital channel for residents to report missed pickups or issues
- Difficulty coordinating on-demand hauling requests for businesses
- Poor data availability for planning and performance monitoring

1.2 Proposed Solution

Bazoor provides a centralized, role-based digital platform that addresses these challenges through four core capabilities: schedule management, real-time tracking, digital reporting, and on-demand hauling coordination. The platform serves government agencies, private haulers, businesses, and residents from a single ecosystem.

1.3 Target Users

- Local Government Units (LGUs) / MENRO — municipal waste schedule and fleet oversight
- Private Hauling Organizations — on-demand service operations and eco-aide management
- Businesses — subscription-based on-demand hauling requests
- Residents — schedule access, truck tracking, report submission, and optional hauling requests
- Collectors (Drivers & Eco-Aides) — field operations, route navigation, and stop management

1.4 Platform

Bazoor is built as a PWA accessible via desktop and mobile browsers. No app store installation is required. The platform supports all role types through a single URL with role-based access control.

2. User Roles & Permissions

2.1 Super Admin

Platform-level administrator responsible for onboarding and approving organizations. Manages subscription access for Government Agencies and Private Hauling Organizations.

2.2 Government Agency Admin (e.g., MENRO)

Manages municipal waste collection schedules, truck fleets, eco-aide accounts, and routes. Reviews analytics and receives digital complaints and reports from community users.

2.3 Private Hauling Organization Admin

Manages eco-aide schedules, hauling request assignments, and service operations. Handles on-demand waste collection and fleet activity monitoring.

2.4 Collector — Driver

Field operator responsible for driving garbage trucks along assigned routes. Uses the platform to view daily routes, mark stops complete, and report on-route issues.

2.5 Collector — Eco-Aide

Field worker performing waste collection on assigned routes or hauling tasks. Uses the platform for the same core field operations as Drivers, with an added on-demand hauling queue.

2.6 Resident (Community User)

End consumer accessing collection schedules, tracking trucks, submitting reports, and optionally requesting on-demand waste pickups. Core features are free; on-demand hauling requires payment per transaction or a subscription.

2.7 Business (Commercial User)

Subscribes to the platform to manage recurring or on-demand waste hauling requests. Cannot access scheduling or fleet management tools.

3. Pricing Model

Bazoorra follows a hybrid pricing model combining subscriptions for organizations and pay-per-use for community users.

User Type	Access Model	Notes
Government Agency	Subscription	Access to fleet management, route monitoring, and service coordination tools.
Private Hauling Org	Subscription	Access to eco-aide management, on-demand hauling, and operational tools.
Business	Subscription (required)	Required to request hauling services and manage waste collection.
Resident	Free (core) / Subscription or Pay-per-request (hauling)	Core features are free. On-demand hauling requires payment per transaction or subscription.

4. Feature Scope & Priority

Features are prioritized using the MoSCoW framework derived from the Feature Matrix. The priority definitions are:

- Must-Have — Core MVP features required before release.
- Should-Have — High-value features that are not MVP-blocking.
- Could-Have — Enhancements deferred to a later sprint.

4.1 Feature Summary by Role

User Role	Must-Have	Should-Have	Could-Have	Total Features
Admin (Government / Hauling Org)	20	7	9	36
Collector — Driver	12	6	7	25
Collector — Eco-Aide	12	6	8	26

4.2 Admin Features (Government Agency & Private Hauling Org)

Module	Feature	Description	Priority	Type
Authentication	Login	Authenticate users and grant system access.	Must-Have	Auth
Authentication	Logout	Terminate active user session.	Must-Have	Auth
Authentication	Set Up Password	Activate account using OTP and password.	Must-Have	Auth
Authentication	Update Password	Change account password.	Must-Have	Auth
User Management	Create Eco-Aide Account	Create a new eco-aide account.	Must-Have	CRUD
User Management	Approve Eco-Aide	Approve eco-aide registration request.	Must-Have	Action
Eco-Aide Management	View Eco-Aides	Display all eco-aides and their statuses.	Must-Have	View
Eco-Aide Management	Edit Eco-Aide Profile	Update eco-aide profile details.	Must-Have	CRUD
Fleet Management	Register Truck	Register a garbage truck in the fleet.	Must-Have	CRUD
Fleet Management	Edit Truck	Update truck details.	Must-Have	CRUD
Route Management	Create Route	Create a collection route and schedule.	Must-Have	CRUD
Route Management	Edit Route	Modify route details.	Must-Have	CRUD
Route Management	Assign Eco-Aide To Route	Assign eco-aide to a route.	Must-Have	Action
Route Management	View Route Details	Display complete route information.	Must-Have	View
Route Monitoring	Route Status Monitoring	Monitor route progress and completion in real time.	Must-Have	Real-time
Request Management	View Hauling Requests	View submitted hauling requests.	Must-Have	View
Request Management	Approve Request	Approve hauling request and assign resources.	Must-Have	Action

Module	Feature	Description	Priority	Type
Request Management	Deny Request	Reject hauling request with reason.	Must-Have	Action
Dashboard	Request Queue Overview	View pending and unassigned requests.	Must-Have	View
GPS Tracking	Live Truck Tracking	Track garbage trucks in real time.	Must-Have	Real-time
Dashboard	Eco-Aide Availability Overview	View eco-aide availability statuses.	Should-Have	View
GPS Tracking	Route Map Preview	Display route map and waypoints.	Should-Have	View
Notifications	Compose Notification	Create and send notifications.	Should-Have	Action
Analytics	Collection Analytics	View pickup volume and completion trends.	Should-Have	View
Account Management	Update Profile Information	Edit administrator profile information.	Should-Have	CRUD
Dashboard	Fleet Status Overview	View active, idle and maintenance vehicles.	Should-Have	View
Dashboard	Summary Statistics	Display key operational metrics.	Should-Have	Automated
User Management	Suspend Eco-Aide	Temporarily restrict account access.	Could-Have	Action
User Management	Deactivate Eco-Aide	Permanently deactivate eco-aide account.	Could-Have	Action
GPS Tracking	ETA Monitoring	Estimate collection arrival times.	Could-Have	Real-time
Notifications	Notification History	View previously sent notifications.	Could-Have	View
Analytics	Revenue Analytics	View paid service revenue metrics.	Could-Have	View
Analytics	Export Analytics Data	Export reports for external analysis.	Could-Have	Export
System Configuration	Auto-Assign Routes	Automatically assign routes based on rules.	Could-Have	Automated
System Configuration	Real-Time Tracking Toggle	Enable or disable GPS tracking.	Could-Have	Action

Module	Feature	Description	Priority	Type
System Configuration	Automatic Reports	Generate recurring reports automatically.	Could-Have	Automated

4.3 Collector — Driver Features

Module	Feature	Description	Priority	Type
Authentication	Driver Login	Authenticate driver access.	Must-Have	Auth
Authentication	OTP Verification	Verify recovery/OTP code.	Must-Have	Auth
Security	Update Password	Change account password.	Must-Have	Auth
Dashboard	View Today's Route	Display assigned route information.	Must-Have	View
Task Management	Mark Stop As Complete	Mark active stop completed.	Must-Have	Action
Issue Management	Report Issue At Stop	Create issue report from a stop.	Must-Have	Action
GPS Tracking	View Route Map	Display route map and navigation.	Must-Have	Real-time
Route Management	View Route Stops	Display all route stops.	Must-Have	View
Task Management	View Assigned Tasks	Display assigned tasks.	Must-Have	View
Task Management	View Task Details	Display selected task details.	Must-Have	View
Issue Management	Create Issue Report	Submit operational incident report.	Must-Have	CRUD
Account Management	Logout	End active session.	Must-Have	Auth
Dashboard	View Dashboard Summary	Display queue, completed, and in-progress stops and truck status.	Should-Have	View
Task Management	View Current Stop	Show current collection stop.	Should-Have	View

Module	Feature	Description	Priority	Type
Collection Management	View Collections Summary	View collection metrics.	Should-Have	View
Issue Management	Upload Issue Photo	Attach issue evidence photos.	Should-Have	CRUD
Issue Management	View Report History	Display submitted reports.	Should-Have	View
Account Management	Update Profile Information	Edit personal information.	Should-Have	CRUD
Authentication	Forgot Password	Initiate password recovery.	Could-Have	Auth
GPS Tracking	Turn-by-Turn Navigation	Provide navigation instructions.	Could-Have	Real-time
Reporting	Export Collection Log CSV	Export collection data as CSV.	Could-Have	Export
Messaging	View Message Channels	Display available conversations.	Could-Have	View
Messaging	View Chat History	Display conversation history.	Could-Have	View
Notifications	Manage Notification Preferences	Configure notification settings.	Could-Have	CRUD
System Configuration	Dark Mode	Toggle dark theme.	Could-Have	Action

4.4 Collector — Eco-Aide Features

Module	Feature	Description	Priority	Type
Authentication	Login	Authenticate eco-aide access.	Must-Have	Auth
Authentication	OTP Verification	Verify recovery/OTP code.	Must-Have	Auth
Security	Update Password	Change account password.	Must-Have	Auth
Dashboard	View Today's Route / Hauling Queue	Display assigned route information.	Must-Have	View
Task Management	Mark Stop As Complete	Mark active stop completed.	Must-Have	Action
Issue Management	Report Issue At Stop	Create issue report from a stop.	Must-Have	Action
GPS Tracking	View Route Map	Display route map and navigation.	Must-Have	Real-time
Route Management	View Route Stops	Display all route stops.	Must-Have	View
Task Management	View Assigned Tasks	Display assigned tasks.	Must-Have	View
Task Management	View Task Details	Display selected task details.	Must-Have	View
Issue Management	Create Issue Report	Submit operational incident report.	Must-Have	CRUD
Account Management	Logout	End active session.	Must-Have	Auth
Dashboard	View Dashboard Summary	Display queue, completed, and in-progress stops and truck status.	Should-Have	View
Task Management	View Current Stop	Show current collection stop.	Should-Have	View

Module	Feature	Description	Priority	Type
Collection Management	View Collections Summary	View collection metrics.	Should-Have	View
Issue Management	Upload Issue Photo	Attach issue evidence photos.	Should-Have	CRUD
Issue Management	View Report History	Display submitted reports.	Should-Have	View
Account Management	Update Profile Information	Edit personal information.	Should-Have	CRUD
Authentication	Forgot Password	Initiate password recovery.	Could-Have	Auth
GPS Tracking	Turn-by-Turn Navigation	Provide navigation instructions.	Could-Have	Real-time
Reporting	Export Collection Log CSV	Export collection data as CSV.	Could-Have	Export
Messaging	View Message Channels	Display available conversations.	Could-Have	View
Messaging	View Chat History	Display conversation history.	Could-Have	View
Notifications	View Notifications List	Display notification feed.	Could-Have	View
Notifications	Manage Notification Preferences	Configure notification settings.	Could-Have	CRUD
System Configuration	Dark Mode	Toggle dark theme.	Could-Have	Action

5. Key System Modules

5.1 Authentication & Role Management

All users authenticate via email and password. New accounts set passwords using a one-time OTP flow. Role-based access control gates each section of the platform to the appropriate user type. Password recovery is supported via OTP email verification.

5.2 Schedule & Route Management

Admins create and manage collection routes with associated stops, schedules, assigned eco-aides, and trucks. Routes can be edited, and eco-aides can be reassigned. Real-time route status monitoring shows live progress as stops are completed in the field.

5.3 Real-Time GPS Tracking

GPS coordinates from trucks are broadcast and displayed on an interactive map for admin oversight and real-time monitoring. Field users (drivers and eco-aides) view their own assigned route map with navigation support. ETA monitoring and turn-by-turn navigation are planned as Could-Have enhancements.

5.4 On-Demand Hauling Requests

Businesses and residents can submit on-demand waste pickup requests. Admins review the request queue, approve or deny submissions with optional reasons, and assign resources (eco-aide and route). Approved requests are surfaced in eco-aide task queues for execution.

5.5 Fleet Management

Admins register trucks with plate numbers, model, and capacity. Trucks are assigned to routes and eco-aides. A fleet status dashboard shows active, idle, and maintenance states. Truck details can be updated as fleet conditions change.

5.6 Eco-Aide & Workforce Management

Admins create and manage eco-aide accounts, including approval of new registrations, profile editing, and account suspension or deactivation. An availability overview allows admins to see which eco-aides are available for scheduling.

5.7 Incident & Issue Reporting

Field users (drivers and eco-aides) can report issues directly from a route stop, including photo attachments. A report history view tracks previously submitted incidents. This module feeds into admin oversight and complaint resolution.

5.8 Notifications & Communications

Admins can compose and send notifications to targeted audiences. Users receive push and/or email notifications for scheduled collection reminders, request status updates, and operational alerts. Notification preferences are configurable per user.

5.9 Analytics & Reporting

A dashboard surface shows key operational metrics including pickup completion rates, fleet status, and request queue volumes. Collection analytics visualize trends over time. Revenue analytics track paid service performance. Reports can be exported for external use.

6. Non-Functional Requirements

6.1 Platform

- Must function as a PWA accessible from modern desktop and mobile browsers.
- No native app installation required. Responsive design for both screen sizes.

6.2 Performance

- Real-time GPS updates should reflect within 5 seconds of field device transmission.
- Dashboard and list views should load within 3 seconds under normal network conditions.

6.3 Security

- All sessions must be authenticated and role-scoped.
- OTP-based account activation and password recovery.
- Sensitive data (personal information, location) must be encrypted at rest and in transit.

6.4 Scalability

- The system should support multiple LGUs and private hauling organizations under a single platform instance.
- Feature access is gated per organization subscription to allow multi-tenant operation.

6.5 Offline / Connectivity

- Field user views (route stops, current stop) should gracefully handle intermittent connectivity.
- Core field actions (marking stops complete, creating reports) should queue locally and sync when online.

7. Out of Scope (v1.0)

The following items are noted in the Feature Matrix as Could-Have enhancements and are explicitly deferred from the initial release:

- Illegal Dumping Hotspot Detection — automated recurring location flagging
- Barangay Sustainability Scorecard & Leaderboard — environmental performance rankings
- Recyclable Exchange Network — material matching and collection facilitation
- Municipal Carbon & Environmental Impact Dashboard — emissions and waste impact tracking
- Overflow Bin Alert — public waste bin overflow reporting
- Waste Collection Service Rating — resident feedback collection
- In-platform messaging and chat between field users
- Dark mode and other UI personalization settings

8. Glossary

Term	Definition
Eco-Aide	Field worker who performs waste collection on foot, on assigned routes or hauling tasks.
LGU	Local Government Unit — the municipal body responsible for public waste management (e.g., MENRO).
MENRO	Municipal Environment and Natural Resources Office — government agency managing environmental services.
MoSCoW	Prioritization framework: Must-Have, Should-Have, Could-Have, Won't-Have.
On-Demand Hauling	A waste pickup request submitted outside of the regular scheduled collection cycle.
PWA	Progressive Web Application — a web app designed to behave like a native mobile app without requiring installation.
Route	A defined collection path with ordered stops, an assigned eco-aide, and an assigned truck.
Stop	A single pickup location within a collection route.
Super Admin	Platform-level administrator who approves and manages all organizations using Bazoora.

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